Information Fusion 126 (2026) 103599

Contents lists available at ScienceDirect

Information Fusion

journal homepage: www.elsevier.com/locate/inffus

Full length article

AI Agents vs. Agentic AI: A Conceptual taxonomy, applications and challenges

Ranjan Sapkota [a],*, Konstantinos I. Roumeliotis [b], Manoj Karkee [a],*

[a] Cornell University, Department of Biological and Environmental Engineering, Ithaca, NY, 14850, USA

[b] University of the Peloponnese, Department of Informatics and Telecommunications, Tripoli, Greece

ARTICLE INFO

Keywords:
AI agents
Agentic AI
Context awareness
Multi-agent systems
Conceptual taxonomy

ABSTRACT

Information fusion, in the context of the Generative AI era, must distinguish AI Agents from Agentic AI. This review critically distinguishes between AI Agents and Agentic AI, offering a structured, conceptual taxonomy, application mapping, and analysis of opportunities and challenges to clarify their divergent design philosophies and capabilities. We begin by outlining the search strategy and foundational definitions, characterizing AI Agents as modular systems driven and enabled by LLMs and LIMs for task-specific automation. Generative AI is positioned as a precursor providing the foundation, with AI agents advancing through tool integration, prompt engineering, and reasoning enhancements. We then characterize Agentic AI systems, which, in contrast to AI Agents, represent a paradigm shift marked by multi-agent collaboration, dynamic task decomposition, persistent memory, and coordinated autonomy. Through a chronological evaluation of architectural evolution, operational mechanisms, interaction styles, and autonomy levels, we present a comparative analysis across both AI agents and agentic AI paradigms. Application domains enabled by AI Agents such as customer support, scheduling, and data summarization are then contrasted with Agentic AI deployments in research automation, robotic coordination, and medical decision support. We further examine unique challenges in each paradigm including hallucination, brittleness, emergent behavior, and coordination failure, and propose targeted solutions such as ReAct loops, retrieval-augmented generation (RAG), automation coordination layers, and causal modeling. This work aims to provide a roadmap for developing robust, scalable, and explainable AI-driven systems.

1. Introduction

Prior to the widespread adoption of AI Agents and Agentic AI (Fig. 1) around 2022 (Before ChatGPT was introduced), the development of autonomous and intelligent agents was deeply rooted in foundational paradigms of artificial intelligence, particularly multi-agent systems (MAS) and expert systems, which emphasized social action and distributed intelligence [1,2]. Notably, Castelfranchi [3] laid the critical groundwork by introducing ontological categories for social action, structure, and mind, arguing that sociality emerges from individual agents' actions and cognitive processes in a shared environment, with concepts like goal delegation and adoption forming the basis for cooperation and organizational behavior. Similarly, Ferber [4] provided a comprehensive framework for MAS, defining agents as entities with autonomy, perception, and communication capabilities, and highlighting their applications in distributed problem-solving, collaborative robotics, and synthetic world simulations.

These early studies established that individual social actions and cognitive architectures are fundamental to modeling collective phenomena, setting the stage for modern AI Agents. This paper builds on these foundational concepts to explore how social action modeling, as proposed in [3,4], informs the design of AI Agents capable of complex, socially intelligent interactions in dynamic environments.

Classical Agent-like systems were designed to perform specific tasks with predefined rules, which offered limited autonomy, and minimal adaptability to dynamic environments. These systems were primarily reactive or deliberative, relying on symbolic reasoning, rule-based logic, or scripted behaviors rather than the learning-driven, context-aware capabilities of modern AI Agents [5,6]. For instance, expert systems used knowledge bases and inference engines to emulate human decision-making in domains like medical diagnosis (e.g., MYCIN [7]). Other notable examples include DENDRAL [8], an expert system for molecular structure prediction; XCON [9], used for computer system

* Corresponding authors.
E-mail addresses: rs2672@cornell.edu (R. Sapkota), mk2684@cornell.edu (M. Karkee).

https://doi.org/10.1016/j.inffus.2025.103599
Received 29 May 2025; Received in revised form 25 July 2025; Accepted 30 July 2025
Available online 22 August 2025

configuration; and CLIPS [10], a rule-based production system framework. Systems like SOAR [11] and the subsumption architecture [12] extended symbolic and reactive logic into cognitive modeling and robotics.

In addition to task-specific reasoning, these agents supported limited forms of social interaction. Early conversational systems like ELIZA [13] and PARRY [14] simulated basic dialogue through pattern matching and script-based responses but lacked genuine understanding or contextual adaptation. Similarly, reactive agents in robotics executed sense-act cycles based on fixed control rules, as seen in early autonomous platforms like the Stanford Cart [15].

Multi-agent systems facilitated coordination among distributed entities, exemplified by auction-based resource allocation in supply chain management [16–18]. Scripted AI in video games, like NPC behaviors in early RPGs, used predefined decision trees [19]. Furthermore, BDI (Belief-Desire-Intention) architectures enabled goal-directed behavior in software agents, such as those in air traffic control simulations [20,21].

However, across these diverse systems, early AI agents shared common limitations: they lacked self-learning, generative reasoning, and adaptability to unstructured or evolving environments. These shortcomings distinguish them from Agentic AI a recent paradigm that builds on deep learning, reinforcement learning, and foundation models to enable agents with contextual awareness, continuous learning, and emergent autonomy [22].

Recent public, academic and industry interest in AI Agents and Agentic AI reflects this broader transition in system capabilities. As illustrated in Fig. 1, Google Trends data demonstrates a significant rise in global search for both terms following the emergence of large-scale generative models in late 2022. This shift is closely tied to the evolution of agent design from the pre-2022 era, where AI Agents operated in constrained, rule-based environments, to the post-LLM period marked by learning-driven, flexible/adaptive architectures [23–25]. These newer systems enable agents to refine their performance over time and interact autonomously with unstructured, dynamic inputs [26–28]. For instance, while pre-modern expert systems required manual updates to static knowledge bases, modern agents leverage emergent neural architectures to generalize across tasks [25]. The surge in trend activity reflects growing awareness of this technological leap, as researchers and practitioners seek tools that go beyond automation toward autonomy and general-purpose reasoning. Moreover, applications are no longer confined to narrow domains like simulations or logistics, but now extend to broad practical settings demanding real-time reasoning and adaptive control. This momentum, as visualized in Fig. 1, highlights the significance of recent architectural advances in scaling autonomous agents for real-world deployment.

The release of ChatGPT in November 2022 marked a pivotal inflection point in the development and public perception of artificial intelligence, catalyzing a global surge in adoption, investment, and research activity [29]. In the wake of this breakthrough, the AI landscape underwent a rapid transformation, shifting from the use of standalone LLMs toward more autonomous, task-oriented frameworks [30]. This evolution progressed through two major post-generative phases: AI Agents and Agentic AI. Initially, the widespread success of ChatGPT popularized Generative Agents, which are LLM-based systems designed to produce novel outputs such as text, images, and code from user prompts [31,32]. These agents were quickly adopted across applications ranging from conversational assistants (e.g., GitHub Copilot [33]) and content-generation platforms (e.g., Jasper [34]) to creative tools (e.g., Midjourney [35]), revolutionizing domains like digital design, marketing, and software prototyping throughout 2023 and beyond.

Although the term AI Agent was first introduced in 1998 [3], it has since evolved significantly with the rise of generative AI. Building upon this generative foundation, a new class of systems commonly referred to as AI Agents has emerged. These agents enhanced LLMs with capabilities for external tool use (e.g., API-based tools), function calling, and sequential reasoning, enabling them to retrieve real-time information and execute multi-step workflows autonomously [36,37]. Example frameworks such as AutoGPT [38] and BabyAGI (https://github.com/yoheinakajima/babyagi) highlight this transition, showcasing how LLMs could be embedded within feedback loops to dynamically plan, act, and adapt in goal-driven environments [39,40]. By late 2023, the field had advanced further into the realm of Agentic AI complex, multi-agent systems in which specialized agents collaboratively decompose goals, communicate, and coordinate toward shared objectives. In line with this evolution, Google introduced the Agent-to-Agent (A2 A) protocol in 2025 [41], a proposed standard designed to enable seamless interoperability among agents across different frameworks and vendors. The protocol is built around five core principles: embracing agentic capabilities, building on existing standards, securing interactions by default, supporting long-running tasks, and ensuring modality agnosticism. These guidelines aim to lay the groundwork for a responsive, scalable agentic infrastructure.

Architectures such as CrewAI demonstrate how these agentic frameworks can accomplish decision-making across distributed roles, facilitating intelligent behavior in high-stake applications including robotics, logistics management, and adaptive decision-support [42].

As the field progresses from Generative Agents toward increasingly autonomous systems of Agentic AI, it becomes critically important to delineate the technological and conceptual boundaries between AI Agents and Agentic AI. While both paradigms build upon LLMs and extend the capabilities of generative systems, they embody fundamentally different architectures, interaction models, and levels of autonomy. AI Agents are typically designed as single-entity systems that perform goal-directed tasks by utilizing external tools, applying sequential reasoning, and integrating real-time information to complete well-defined functions [25,43]. In contrast, Agentic AI systems are composed of multiple, specialized agents that coordinate, communicate, and dynamically allocate sub-tasks within a broader workflow to achieve a common goal(s) [22,44]. This architectural distinction highlights clear and significant differences in scalability, adaptability, and application scope.

Understanding and formalizing the taxonomy between these two paradigms **(AI Agents and Agentic AI)** is scientifically significant for several reasons. First, it enables more precise system design by aligning computational frameworks with problem complexity ensuring that AI Agents are deployed for modular, tool-assisted tasks, while Agentic AI is employed for orchestrated multi-agent operations. Moreover, it allows for appropriate benchmarking and evaluation: performance metrics, safety protocols, and resource requirements differ substantially between agents designed for carrying out individual tasks and system of distributed agents designed for accomplishing complex, coordinated tasks. Additionally, clear taxonomy reduces development inefficiencies by preventing the misapplication of design principles such as assuming inter-agent collaboration in a system designed for single-agent execution. Without this clarity, developers and practitioners risk both under-engineering complex scenarios that require agentic coordination and over-engineering simple applications that could be solved with a single AI Agent.

This article aims to clarify the differences between AI Agents and Agentic AI, providing researchers with a foundational understanding of these technologies. The objective of this study is to formalize the distinctions, establish a shared vocabulary, and provide a structured taxonomy between AI Agents and Agentic AI that informs the next generation of intelligent agent design across academic and industrial domains, as illustrated in Fig. 2.

This review also provides a comprehensive conceptual and architectural analysis of the progression from traditional AI Agents to emergent Agentic AI systems. Rather than organizing the study around formal research questions common in review articles, we adopt a sequential, layered structure that clearly lays out the historical and technical evolution of these paradigms. Beginning with a detailed description

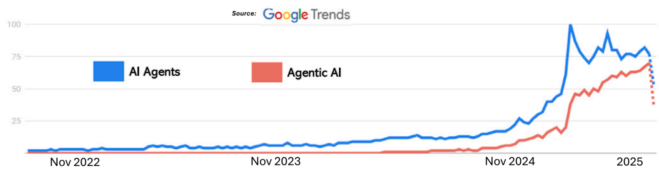

Fig. 1. Global Google search trends showing rising interest in "AI Agents" and "Agentic AI" since November 2022 when the ChatGPT was first introduced.

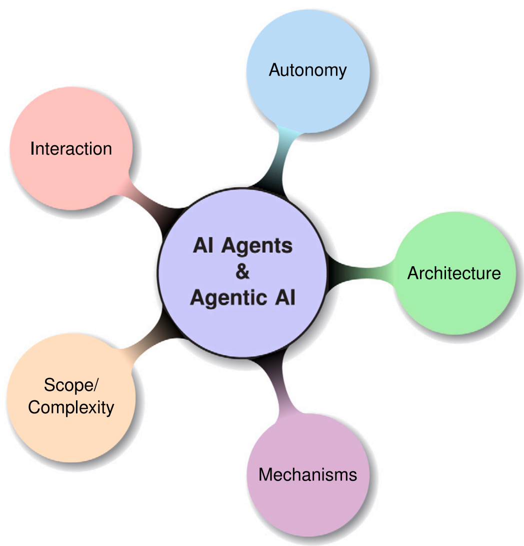

Fig. 2. Mind map of this research for exploring concepts, applications and challenges of AI Agents and Agentic AI. Each color-coded branch represents a key dimension of comparison: Architecture, Mechanisms, Scope/Complexity, Interaction, and Autonomy.

of our search strategy and selection criteria, we first establish the foundational understanding of AI Agents by analyzing their defining attributes, such as autonomy, reactivity, and tool-based execution. We then explore the critical role of foundational models, specifically LLMs and Large Image Models (LIMs), which serve as the core reasoning and perceptual engines that drive agentic behavior. Subsequent sections examine how generative AI systems have served as precursors to more dynamic, interactive agents, setting the stage for the emergence of Agentic AI. Through this perspective, we examine and present the conceptual leap from isolated, single-agent systems to orchestrated multi-agent architectures, highlighting their structural distinctions, coordination strategies, and collaborative mechanisms. We further map the architectural evolution by analyzing the core system components of both AI Agents and Agentic AI, offering comparative description of their planning, memory, orchestration, and execution layers. Building on this foundation, we review application domains spanning customer support, healthcare, research automation tasks, and robotics, while categorizing real-world deployments by system capabilities and coordination complexity. We then assess key challenges faced by both paradigms including hallucination, limited reasoning depth, causality deficits, scalability issues, and governance risks. To address these limitations, we outline opportunities for emerging solutions such as retrieval-augmented generation, tool-based reasoning, memory architectures, and simulation-based planning. The review concludes with a forward-looking roadmap that envisions the convergence of modular AI Agents and orchestrated Agentic AI in mission-critical domains such as autonomous vehicles, finance, and healthcare, and beyond. We aim to provide researchers with a structured taxonomy and actionable insights to guide the design, deployment, and evaluation of next-generation agentic AI systems.

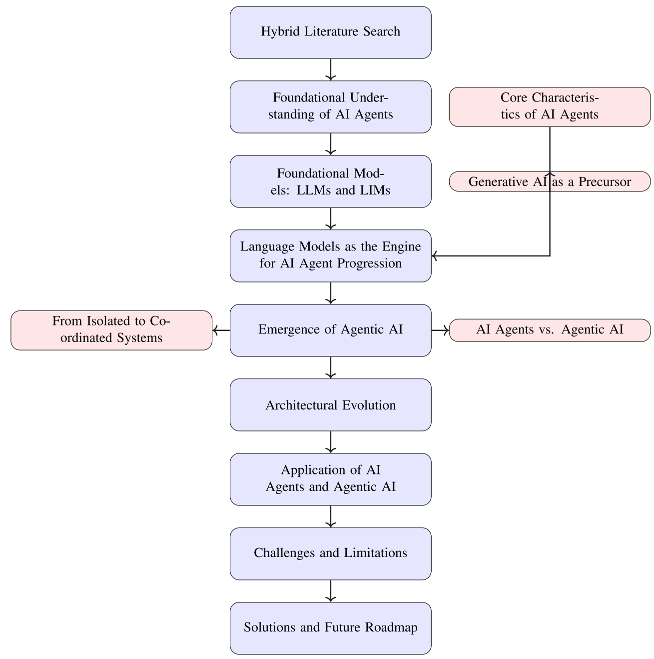

Fig. 3. Methodology pipeline showing the conceptual and architectural progression from foundational AI Agents to emergent Agentic AI. Each stage reflects a distinct layer of conceptual depth, highlighting key mechanisms, comparisons, challenges, and future directions.

Despite growing interest in AI Agents and Agentic AI, existing survey efforts often conflate the two under the broad umbrella of intelligent agents, leading to conceptual ambiguity and misaligned system design. Recent literature either focuses narrowly on tool-augmented LLMs for task automation or broadly discusses multi-agent systems without acknowledging the architectural and functional transition brought forth by Agentic AI. This review offers the first structured taxonomy that formally separates these paradigms across dimensions of autonomy, coordination, interaction, and reasoning scope. By positioning AI Agents as modular, single-entity systems and Agentic AI as orchestrated ecosystems with emergent behaviors, this work addresses a critical gap in distinguishing design principles, deployment strategies, and evaluation criteria. The timing of this distinction is particularly important now, as real-world applications increasingly demand scalable, multi-agent intelligence (e.g., in robotics, healthcare, and scientific discovery), yet developers often lack a clear framework for choosing between isolated agents and collaborative agentic architectures. This review bridges that gap, enabling more precise alignment of system capabilities with task complexity, and informing both research agendas and practical deployments in the post-LLM era.

1.1. Methodology overview

This review adopts a structured, multi-stage methodology designed to capture the evolution, architecture, application, and limitations of AI Agents and Agentic AI. The process is visually summarized in Fig. 3, which delineates the sequential flow of topics and concepts explored in this study. The analytical framework was organized to analyze and present the progression from basic agentic constructs rooted in LLMs to advanced multi-agent orchestration systems. Each step of the review was based on the rigorous synthesis of the literature from across academic sources and AI-powered platforms, enabling a comprehensive understanding of the current landscape and its emerging trends.

The review begins by establishing a *foundational understanding of AI Agents*, examining their core definitions, design principles, and architectural modules as described in the literature. These include components such as perception, reasoning, and action selection, along with early applications like customer service bots and retrieval assistants. This foundational layer serves as the conceptual entry point into the broader agentic paradigm.

Next, we discuss the role of *LLMs as core reasoning components*, emphasizing how pre-trained language models enable modern AI Agents. This section details how LLMs, through instruction fine-tuning and reinforcement learning from human feedback (RLHF), enable natural language interaction, planning, and limited decision-making capabilities. We also identify their limitations, such as hallucinations, static knowledge, and a lack of causal reasoning.

Building on these foundations, the review proceeds to the *emergence of Agentic AI*, which represents a significant conceptual advancement. Here, we highlight the transformation from tool-augmented single-agent systems to collaborative, distributed ecosystems of interacting agents. This shift is driven by the need for systems capable of decomposing goals, assigning subtasks, coordinating outputs, and adapting dynamically to changing contexts, which are the capabilities that surpass what isolated AI Agents can offer.

The next section examines the *architectural evolution from AI Agents to Agentic AI systems*, contrasting simple, modular agent designs with complex orchestration frameworks. We describe enhancements such as persistent memory, meta-agent coordination, multi-agent planning loops (e.g., ReAct and Chain-of-Thought prompting), and semantic communication protocols. Comparative architectural analysis is supported with examples from platforms like AutoGPT, CrewAI, and LangGraph.

Following the architectural exploration, the review presents an in-depth analysis of *application domains* where AI Agents and Agentic AI are being deployed. The paper discusses four representative application areas for each paradigm. For AI Agents, these include Customer Support Automation, Internal Enterprise Search, Email Filtering and Prioritization, and Personalized Content Recommendation. For Agentic AI, the applications span Multi-Agent Research Assistants, Intelligent Robotics Coordination, Collaborative Medical Decision Support, and Adaptive Workflow Automation. These use cases are examined with respect to system complexity, real-time decision-making, and collaborative task execution.

Subsequently, we address the *challenges and limitations* inherent to both paradigms. For AI Agents, we focus on issues like hallucination, prompt brittleness, limited planning ability, and lack of causal understanding. For Agentic AI, we identify higher-order challenges such as inter-agent misalignment, error propagation, unpredictability of emergent behavior, explainability deficits, and adversarial vulnerabilities. These problems are critically examined with references to recent experimental studies and technical reports.

Finally, the review outlines *potential solutions to overcome these challenges*, drawing on recent advances in causal modeling, retrieval-augmented generation (RAG), multi-agent memory frameworks, and robust evaluation pipelines. These strategies are discussed not only as technical fixes but as foundational requirements for scaling agentic systems into high-stakes domains such as healthcare, finance, and autonomous robotics.

In summary, this methodological structure enables a systematic and comprehensive assessment of the state of AI Agents and Agentic AI. By sequencing the analysis from foundational understanding, to model integration, architectural advancements, applications, and to limitations and potential solutions, the study aims to provide both theoretical clarity and practical guidance to researchers and practitioners navigating this rapidly evolving field.

1.1.1. Search strategy

To develop this review, we implemented a hybrid search methodology combining traditional academic repositories and AI-enhanced literature discovery tools. Specifically, twelve platforms were queried: academic databases such as Google Scholar, IEEE Xplore, ACM Digital Library, Scopus, Web of Science, ScienceDirect, and arXiv; and AI-powered interfaces including ChatGPT, Perplexity.ai, DeepSeek, Hugging Face Search, and Grok. Search queries incorporated Boolean combinations of terms such as "AI Agents", "Agentic AI", "LLM Agents", "Tool-augmented LLMs", and "Multi-Agent AI Systems".

Targeted queries such as "Agentic AI + Coordination + Planning", and "AI Agents + Tool Usage + Reasoning" were also employed to retrieve papers addressing both conceptual underpinnings and system-level implementations. Literature inclusion was based on their the significance in terms of novelty, empirical evaluation, architectural contribution, and citation impact. The rising global interest in these technologies, as illustrated in Fig. 1 using Google Trends data, underscores the urgency of synthesizing this emerging knowledge space.

2. Foundational understanding of AI Agents

AI Agents can be defined as autonomous software entities engineered for goal-directed task execution within bounded digital environments [22,45]. These agents are defined by their ability to perceive structured or unstructured inputs [46], to reason over contextual information [47,48], and to initiate actions toward achieving specific objectives, often acting as surrogates for human users or subsystems [49]. Unlike conventional automation scripts, which follow deterministic workflows, AI Agents demonstrate reactive intelligence and some level of adaptability, allowing them to interpret dynamic inputs and reconfigure outputs accordingly [50]. Their adoption has been reported across a wide range of application domains, including customer service automation [51,52], personal productivity assistance [53], organizational information retrieval [54,55], and decision support systems [56,57].

A notable example of autonomous AI agents in Anthropic's "Computer Use" project computer use, which showcases how their Claude model can interact with a computer in much the same way a human would. In this project, Claude is trained to visually interpret what is on a computer screen, control the mouse and keyboard, and navigate through various software applications. This allows Claude to automate repetitive tasks, such as filling out forms or copying data, as well as more complex activities like building and testing software by opening code editors, running commands, and debugging issues. Beyond these structured tasks, Claude can also handle open-ended assignments like conducting online research, gathering and organizing information from multiple sources, and even creating calendar events based on its findings. The key innovation is that Claude operates in an "agent loop", where it receives a goal, decides on the next action, performs that action, observes the result, and repeats this process until the task is complete. This enables Claude to independently use existing computer tools and interfaces to accomplish a wide range of objectives, making it a powerful example of how autonomous AI agents can automate both routine and complex workflows.

2.0.1. Core characteristics of AI Agents

AI Agents are widely conceptualized as instantiated operational instances of artificial intelligence designed to interface with users, software ecosystems, or digital infrastructures to develop goal-directed behavior [58–60]. These agents are different than general-purpose LLMs in the sense that they exhibit structured initialization, bounded autonomy, and persistent task orientation. While LLMs primarily function as reactive prompt followers [61], AI Agents operate automatically within explicitly defined scopes, engaging dynamically with inputs and producing actionable outputs in real-time environments [62].

Fig. 4 illustrates the three foundational characteristics commonly incorporated by architectural taxonomies and empirical deployments of AI Agents. These characteristics include *autonomy*, *task-specificity*, and *reactivity with adaptation*.

Together, these three characteristics provide a foundational framework for understanding and evaluating AI Agents across deployment scenarios. The remainder of this section elaborates on each characteristic, offering theoretical background and illustrative examples.